Solution Architecture

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Project Name	Voice Based Concept Understanding Analyser
Maximum Marks	5 Marks

Solution Architecture

The Voice-Based Concept Understanding Analyser solution architecture defines the overall structure of the system and explains how different components work together to evaluate a student's conceptual understanding through speech. It bridges the gap between the educational problem and an AI-based technology solution by integrating Speech-to-Text, Natural Language Processing (NLP), Generative AI, and reporting modules.

The solution architecture aims to:

- Develop an AI-powered system that evaluates students' conceptual understanding from spoken explanations.
- Convert voice input into text using a Speech-to-Text engine for further processing.
- Analyze the transcript using NLP and Generative AI models to understand the concept explained by the student.
- Compare the student's explanation with the expected concept to measure accuracy, clarity, completeness, and confidence.
- Generate personalized feedback, strengths, weaknesses, and improvement suggestions.
- Display the evaluation results through an interactive dashboard with transcript, score, audio visualization, and downloadable PDF report.
- Store transcripts, evaluation results, and reports securely for future reference.
- Provide a scalable, reliable, and user-friendly architecture that supports continuous learning and performance assessment.

The solution architecture also defines the major system components, their interactions, data flow, functional requirements, and implementation phases to ensure efficient development, deployment, and maintenance of the application.

Voice Based Concept Understanding Analyser – Solution Architecture Diagram